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Dietary Restrictions, Physical Health, and Cardiovascular Risk: Mechanisms, Clinical Evidence, and Safety Considerations: A Narrative Review

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Abstract

Background. Intermittent fasting (IF) has emerged as a prominent non-pharmacological strategy in lifestyle medicine and sports science, particularly in the context of cardiovascular disease (CVD) prevention and cardiometabolic optimization. Growing interest in IF reflects its potential to target cardiometabolic risk factors through time-dependent metabolic adaptations rather than continuous caloric restriction.

Methods. This narrative review synthesizes current evidence derived from basic experimental research, clinical trials, and systematic reviews evaluating the cardiovascular, metabolic, and physical performance effects of intermittent fasting. The analysis focuses on three primary fasting regimens: time-restricted eating (TRE), alternate-day fasting (ADF), and modified fasting regimens (MFR).

Results. Evidence demonstrates that IF induces metabolic switching from glucose utilization to fatty acid oxidation and ketone body production. These adaptations correlate with improvements in blood pressure, atherogenic lipid profiles, glucose–insulin homeostasis, and oxidative stress pathways. Mechanistic studies suggest enhanced autophagy, improved mitochondrial biogenesis, and circadian rhythm alignment contribute to cardiometabolic cardioprotection. However, data regarding long-term safety, hard cardiovascular endpoints, and potential risks such as transient energy deficits remain critical areas of ongoing evaluation.

Conclusions. Intermittent fasting represents a highly effective lifestyle intervention for enhancing cardiometabolic health and reducing cardiovascular risk. While short- and medium-term outcomes are favorable, structured clinical oversight is necessary to ensure safety and long-term compliance in clinical and active populations.

Keywords: intermittent fasting, cardiovascular disease, cardiometabolic health, circadian rhythm, physical health, Dietary Restrictions

1. Introduction

Cardiovascular diseases (CVDs) remain the undisputed leading cause of morbidity and premature mortality globally, posing a structural, financial, and clinical strain on modern healthcare systems. The classical pathogenesis of CVDs-encompassing coronary artery disease, heart failure, and cerebrovascular accidents-is tightly linked to progressive, systemic metabolic dysfunction. This dysfunction is primarily driven by chronic positive energy balance and modern sedentary lifestyles, which facilitate the expansion of dysfunctional visceral adipose tissue, systemic low-grade inflammation, microvascular endothelial dysfunction, and progressive peripheral insulin resistance. In the face of the global epidemic of metabolic syndrome, contemporary cardiology is actively shifting its paradigm toward integrative primary prevention. There is an urgent medical need for non-pharmacological interventions that are not only biologically effective, but also sustainable and acceptable for long-term population-wide adoption without triggering high therapeutic drop-out rates or excessive reliance on early polypharmacy.

Intermittent fasting (IF) has risen to prominence as an attractive, biologically plausible dietary paradigm within translational lifestyle medicine and preventive sports science [1-3]. Defined broadly, IF comprises eating patterns in which individuals undergo recurring, timed periods of voluntary abstinence from caloric intake, alternating with periods of standard or ad libitum food consumption [1,4]. In contrast to traditional continuous caloric restriction (CCR)-which focuses almost exclusively on daily quantitative energy reduction and often triggers counter-regulatory metabolic slowing-IF centers on the temporal distribution and circadian alignment of food intake. This operational shift harnesses evolutionarily conserved genomic and biochemical adaptation pathways that originally evolved to maintain cellular viability during prolonged starvation [1,4,5]. By oscillating between nutrient-abundant and nutrient-scarce states, IF activates endogenous biological cascades, inducing cellular repair mechanisms, suppressing pro-inflammatory signaling, enhancing physical performance metrics, and increasing overall metabolic flexibility [2,3,6].

At the physiological core of the health benefits triggered by IF lies the systemic phenomenon known as "metabolic switching" [1,2]. Under standard conditions of uninterrupted food availability and frequent postprandial states, human cardiac, vascular, and skeletal tissues rely predominantly on exogenous glucose and hepatic glycogen storage for adenosine triphosphate (ATP) synthesis. However, when a fasting period extends beyond approximately 10 to 16 hours, glycogen stores in the liver become depleted, forcing a systemic metabolic transition toward adipose tissue lipolysis and hepatic fatty acid beta-oxidation [1,2,7]. This metabolic

switch results in the robust hepatic synthesis of ketone bodies-predominantly Beta-hydroxybutyrate (β HB) and acetoacetate [1,2]. Far from being mere metabolic byproducts, ketone bodies serve as an energetically dense, highly efficient metabolic fuel for myocardium and vascular endothelial cells, while simultaneously acting as signaling molecules that suppress oxidative stress, inhibit the NLRP3 inflammasome, and promote anti-atherogenic gene expression [1,2,7].

2. Aim of the Study

The primary objective of this narrative review is to synthesize and critically analyze the evolving body of scientific evidence regarding the impact of various intermittent fasting regimens on cardiovascular health, cardiometabolic risk parameters, and athletic performance. Specifically, this paper aims to:

1. Deconstruct the primary physiological and cellular mechanisms underlying IF-induced cardioprotection, including autophagy activation, mitochondrial biogenesis, anti-inflammatory cascades, and circadian gene expression.
2. Evaluate clinical trial findings concerning the impact of different fasting protocols on classic cardiovascular risk factors (blood pressure, atherogenic lipid profiles, and insulin resistance).
3. Thoroughly discuss the safety, tolerability, clinical limitations, and practical implementation barriers of IF, particularly in high-risk groups such as patients with type 2 diabetes or active individuals.

3. Classification and Characteristics of Intermittent Fasting Protocols

Establishing precise nomenclature and functional taxonomies for intermittent fasting protocols is critical for interpreting physiological outcomes, athletic performance maintenance, therapeutic compliance, and clinical safety [1-3,5,8].

3.1. Alternate-Day Fasting (ADF)

Alternate-day fasting involves a strict, alternating schedule between "feasting days" (ad libitum consumption) and "fasting days." In its classical form, fasting days involve complete total caloric deprivation for 24 hours. However, most contemporary clinical trials utilize modified ADF, allowing the consumption of approximately 20–25% of baseline daily energy requirements (roughly 500–600 kcal) on fasting days, usually consumed in a single meal.

Clinical evaluations demonstrate that ADF produces rapid and marked reductions in overall body mass, visceral fat volume, circulating triglycerides, and atherogenic ApoB-containing

lipoproteins [1,3,9]. Despite these impressive metabolic outcomes, ADF faces substantial compliance barriers in real-world outpatient settings. Participants frequently report persistent hyperphagia on non-fasting days, elevated hunger scores, irritability, and social disruption, contributing to higher attrition rates during extended intervention periods beyond 12 weeks [1,9,10].

3.2. Modified Fasting Regimens (MFR)

Modified fasting regimens incorporate intermittent, non-consecutive days of severe caloric restriction within a standard weekly cycle, rather than an alternating daily schedule. The most widely adopted and clinically validated MFR model is the 5:2 diet protocol. In this regimen, individuals consume their regular diet for five days of the week and undergo severe caloric restriction (typically 500–600 kcal/day) on two non-consecutive days.

Evidence indicates that MFR induces clinically meaningful weight loss and adipose tissue loss comparable to continuous caloric restriction [1,3,9]. Mechanistic studies suggest that MFR may confer superior improvements in peripheral tissue insulin sensitivity and hepatic fat content compared to equivalent continuous caloric restriction models, likely due to the periodic deep depletion of intrahepatic lipid pools [1,3,11].

3.3. Time-Restricted Eating (TRE)

Time-restricted eating limits daily energy intake to a restricted temporal window-typically lasting between 4 to 10 hours-enforcing a consistent 14- to 20-hour daily fast without requiring explicit daily calorie counting [1-3,12]. Common variations include 16:8 (16 hours fasting, 8 hours eating window) and 18:6 protocols.

From a patient-centered and athletic perspective, TRE consistently demonstrates high rates of long-term adherence and satisfaction due to its simplicity and social compatibility [2,12,13]. A crucial physiological feature determining the efficacy of TRE is its alignment with central and peripheral circadian clocks located in cardiac, vascular, hepatic, and muscle tissues [1,4]. Consuming meals in synchronization with physiological circadian rhythms (e.g., early time-restricted eating, where food intake is aligned with peak daylight hours and early-day insulin sensitivity) maximizes cardiometabolic benefits, whereas late-evening eating disrupts circadian clock gene alignment, promoting metabolic endotoxemia and insulin resistance [1,13,14].

4. Molecular Mechanisms of Cardioprotection

The biological mechanisms through which intermittent fasting exerts its systemic cardioprotective effects involve a tightly coordinated network of cellular energy sensing, metabolic substrate shifting, cellular clearance, and anti-inflammatory pathways.

When caloric intake is temporarily restricted, the human organism undergoes a foundational transition at both the systemic tissue level and cellular level:

1. **Systemic Energy Transition via Hepatic Ketogenesis:** Extended fasting leads to the progressive depletion of intrahepatic glycogen stores. This energy deficit triggers adipose tissue lipolysis, mobilizing free fatty acids that undergo hepatic β -oxidation. Consequently, the liver produces ketone bodies-most prominently β -hydroxybutyrate (β HB). Beyond functioning as an energetically dense, highly efficient energy substrate for cardiac myocytes and endothelial cells, β HB acts as an active signaling molecule that inhibits the NLRP3 inflammasome, suppresses inflammatory cytokines (such as hs-CRP and IL-6), and mitigates vascular oxidative stress.
2. **Cellular Energy Sensing via Nutrients Modulation:** At the intracellular level, nutrient deprivation elevates the cellular AMP-to-ATP ratio, directly activating 5'-AMP-activated protein kinase (AMPK)-the primary intracellular energy sensor. Concurrently, drops in circulating glucose, insulin, and amino acids cause the functional inactivation of the Mechanistic Target of Rapamycin Complex 1 (mTORC1) pathway.
3. **Autophagy and Mitophagy Activation:** The synergistic effect of AMPK upregulation and mTORC1 suppression potently stimulates macroautophagy and selective mitophagy in ventricular cardiomyocytes and vascular endothelial cells. This cellular housecleaning process drives the enzymatic degradation and clearance of misfolded proteins, dysfunctional sarcoplasmic reticulum elements, and oxidized lipids.
4. **Mitochondrial Biogenesis and Endothelial Repair:** Fasting activates Sirtuin pathways (SIRT1/SIRT3) and Peroxisome proliferator-activated receptor gamma coactivator 1-alpha (PGC-1 α). This upregulates the synthesis of functional mitochondria (mitochondrial biogenesis) and activates Nuclear factor erythroid 2-related factor 2 (Nrf2), which promotes the expression of endogenous antioxidant enzymes (superoxide dismutase, catalase).

Together, these integrated pathways enhance myocardial energy efficiency, maintain microvascular junctional integrity, reduce myocardial fibrosis, and improve overall cardiac ejection fraction [1-4,15-18].

4.1. Autophagy and Mitochondrial Biogenesis

At the cellular level, intermittent fasting triggers intracellular clearance pathways essential for maintaining cardiac function and structural integrity [1,4]. Fasting alters intracellular energy ratios-specifically increasing the AMP-to-ATP ratio-which directly activates 5'-AMP-activated protein kinase (AMPK), the master cellular energy sensor [1,15]. Concurrently, the systemic drop in circulating insulin and amino acid levels inactivates the Mechanistic Target of Rapamycin Complex 1 (mTORC1) pathway [1,16].

The suppression of mTORC1 combined with AMPK activation potently upregulates macroautophagy in ventricular cardiomyocytes and vascular endothelial cells [1]. This upregulation facilitates the clearance of misfolded protein aggregates, damaged sarcoplasmic reticulum components, and dysfunctional, reactive oxygen species (ROS)-producing mitochondria (mitophagy) [1,15]. By clearing damaged mitochondrial populations and activating Peroxisome proliferator-activated receptor gamma coactivator 1-alpha (PGC-1 α), IF stimulates mitochondrial biogenesis, enhancing myocardial ATP generation and preventing adverse cardiac remodeling [1,16].

4.2. Modulation of Inflammation and Oxidative Stress

Chronic systemic low-grade inflammation is a primary driver of atherogenesis, plaque instability, and heart failure. Intermittent fasting exerts multi-factorial anti-inflammatory and antioxidative effects across the cardiovascular network [1,3,17]. Fasting significantly downregulates systemic circulating concentrations of pro-inflammatory cytokines, including high-sensitivity C-reactive protein (hs-CRP), interleukin-6 (IL-6), and tumor necrosis factor-alpha (TNF- α) [1,3,17].

Elevated circulating levels of β -hydroxybutyrate (β HB) directly bind to and inhibit the assembly of the NLRP3 inflammasome within circulating monocytes and vascular macrophages [1,7]. On an intracellular level, the energetic stress induced by IF activates the sirtuin pathway (SIRT1/SIRT3) and nuclear factor erythroid 2-related factor 2 (Nrf2) [1,18]. Activated Nrf2 translocates to the nucleus to induce the transcription of key antioxidant enzymes-such as superoxide dismutase (SOD), catalase, and glutathione peroxidase-thereby neutralizing free radicals, protecting vascular endothelial junctional proteins, and increasing nitric oxide (NO) bioavailability [1,18].

5. Effects on Cardiovascular Risk Factors

5.1. Regulation of Blood Pressure

Clinical studies evaluating time-restricted eating and alternate-day fasting protocols consistently demonstrate statistically and clinically significant reductions in both systolic (ranging from 3 to 10 mmHg) and diastolic (ranging from 2 to 6 mmHg) blood pressure [12,19]. These antihypertensive effects are mediated through multiple physiological pathways:

- **Sympatholytic Shift:** Fasting reduces tonic sympathetic nervous system outflow to the heart and kidneys, partially mediated by decreased basal circulating insulin levels.
- **Natriuresis:** Reduced renal tubular sodium reabsorption occurs secondary to suppressed hyperinsulinemia.
- **Vascular Reactivity:** Upregulated endothelial nitric oxide synthase (eNOS) activity enhances vascular compliance, decreases arterial stiffness, and improves flow-mediated dilation (FMD) [12,19].

5.2. Lipid Profile and Atherogenesis

Intermittent fasting protocols exert beneficial modulations across complex lipid subfractions [1,3,13,20]. Clinical trial data show that regular IF adherence causes notable reductions in total circulating fasting triglycerides (frequently reduced by 15% to 30%) and low-density lipoprotein cholesterol (LDL-C) [1,3,13,20]. The primary physiological mechanism driving triglyceride lowering is the suppression of hepatic VLDL secretion combined with enhanced peripheral lipoprotein lipase (LPL) activity during fasting windows [3,13].

Crucially, IF favorably alters lipoprotein particle morphology and atherogenicity [1,13,21]. Fasting regimens promote a shift away from small, dense, highly oxidation-prone LDL particles (sdLDL) toward larger, less atherogenic, buoyant LDL particles [13,21]. Concurrently, studies document modest increases or functional shifts in high-density lipoprotein cholesterol (HDL-C) particles, improving reverse cholesterol transport capacity from arterial macrophages [13,21].

5.3. Glucose Homeostasis and Systemic Cardiometabolic Adaptations

Insulin resistance and hyperinsulinemia are primary drivers of cardiovascular damage, accelerating coronary plaque calcification and myocardial interstitial fibrosis. Intermittent fasting protocols demonstrate high potency in restoring systemic glucose-insulin homeostasis, evidenced by significant drops in fasting blood glucose, glycated hemoglobin (HbA1c), basal fasting insulin, and the HOMA-IR index [1,3,13,22].

By extending the physiological duration of low circulating insulin states, IF enhances insulin receptor substrate (IRS-1) signaling and promotes GLUT4 translocation in skeletal muscle and adipose tissue upon nutrient reintroduction [3,13,22]. Moreover, the cyclic depletion of intrahepatic lipid content via beta-oxidation alleviates hepatic insulin resistance, thereby restoring baseline suppression of endogenous gluconeogenesis during non-eating hours [1,13,22].

The multi-systemic physiological mechanisms and observed clinical outcomes across core cardiometabolic parameters can be summarized through the following key pathways:

- **Blood Pressure Regulation:** Driven primarily by a reduction in tonic sympathetic outflow, enhanced endothelial nitric oxide synthase (eNOS) activity, and increased renal sodium excretion (natriuresis). These physiological adaptations yield clinical reductions in both systolic (3 to 10 mmHg) and diastolic (2 to 6 mmHg) blood pressure.
- **Triglyceride Mobilization:** Induced via the suppression of hepatic VLDL synthesis coupled with elevated peripheral lipoprotein lipase (LPL) clearance during fasting windows, resulting in a marked circulating triglyceride decrease ranging from 15% to 30%.
- **Lipoprotein Particle Remodeling:** Mediated through the structural modification of low-density lipoproteins, shifting the distribution away from small, dense, oxidation-prone LDL (sdLDL) toward large, buoyant LDL subfractions, thereby significantly dampening vascular atherogenic potential.
- **Insulin Sensitivity Restoration:** Achieved through the activation of muscle IRS-1 signaling, hepatic lipid clearing, and prolonged periods of low baseline insulin levels, leading to clear clinical reductions in HOMA-IR scores, glycated hemoglobin (HbA1c), and fasting glucose.
- **Inflammatory Cascade Attenuation:** Triggered by the direct inhibition of the NLRP3 inflammasome by circulating β -hydroxybutyrate (β HB) alongside SIRT1/Nrf2 nuclear pathway activation, producing measurable declines in systemic biomarkers such as hs-CRP, IL-6, and vascular oxidative stress.

Meta-analyses highlight that while the overarching cardiometabolic trends of IF are positive, the magnitude of clinical benefits varies based on baseline patient characteristics, intervention duration, underlying athletic or metabolic status, and precise fasting compliance [1-3,23,24]. Standardized clinical protocols remain necessary to minimize outcome heterogeneity across prospective investigations [23,25].

6. Safety Considerations and Clinical Limitations

While intermittent fasting exhibits widespread metabolic benefits, it is not without physiological risks and clinical contraindications. Safe implementation requires careful risk stratification and individualized clinical oversight [1,3,14,26].

A primary clinical concern involves patients with Type 1 or Type 2 Diabetes Mellitus managed with exogenous insulin or insulin secretagogues (e.g., sulfonylureas). In these cohorts, prolonged fasting windows dramatically elevate the incidence of severe hypoglycemia unless baseline pharmacological dosages are preemptively adjusted [1,3,14,26]. Furthermore, individuals with chronic kidney disease, advanced heart failure taking high-dose diuretics, or orthostatic hypotension are prone to acute volume depletion and electrolyte imbalances during extended fasting intervals [14,26].

In physically active individuals and athletes, adaptation to IF may initially induce transient decreases in energy levels, power output, or subjective fatigue during high-intensity training sessions until full metabolic keto-adaptation occurs [2]. Additionally, serious epidemiological questions have been raised regarding extreme eating windows shorter than 8 hours, emphasizing that aggressive caloric deprivation does not linearly scale with clinical safety [14,27]. Finally, IF is strictly contraindicated in individuals with a history of eating disorders, during pregnancy and lactation, and in frail elderly populations prone to sarcopenia [26,27].

7. Discussion

The expanding body of scientific evidence supports intermittent fasting as a valuable, non-pharmacological component of preventive lifestyle medicine and health promotion [1,2,13]. However, successfully translating these findings into routine clinical practice presents notable structural challenges and unresolved scientific questions.

A key translational topic centers on determining the optimal daily time window for TRE protocols. Evidence increasingly favors **early time-restricted eating (eTRE)**-where daily nutrient consumption begins at waking and concludes in the early-to-mid afternoon-over late time-restricted eating patterns [13,14]. eTRE aligns food intake with natural diurnal peaks in gastrointestinal motility, insulin sensitivity, pancreatic beta-cell responsiveness, and diet-induced thermogenesis. Conversely, delaying nutrient intake into the late evening forces metabolic processing to occur during a period of physiological nocturnal insulin resistance, blunting the cardiometabolic benefits [14,28].

Beyond biochemical surrogate endpoints, the impact of intermittent fasting on long-term **patient-reported outcomes (PROMs)**, athletic sustainability, and **health-related quality of**

life (HRQoL) demands rigorous evaluation [1,2,29]. Evidence from sports psychology, behavioral medicine, and physical education indicates that improvements in objective clinical biomarkers do not automatically translate into improved subjective well-being if the dietary intervention imposes severe social isolation, chronic hunger, fatigue, or psychological stress [2,3,6,29,30]. Dietary protocols must fit within an individual's cultural, occupational, and psychological environment to remain long-term viable interventions [10,30].

A major limitation of intermittent fasting remains the decay in behavioral adherence over extended time horizons. While short-term clinical trials display high compliance rates, long-term observational follow-ups reveal a progressive decline in adherence after 6 to 12 months. Individuals frequently revert to baseline habits due to environmental cues, social obligations, and hedonic hunger drives, resulting in weight regain and the reversal of cardiometabolic improvements [1,9,10].

Finally, major cardiovascular societies remain cautious regarding the formal endorsement of IF as a standalone therapeutic strategy equal to validated pharmacological options or traditional Mediterranean-style diets. This conservatism stems from the lack of multi-center randomized controlled trials evaluating **hard clinical cardiovascular outcomes**-such as major adverse cardiovascular events (MACE) and mortality [23,28]. Future clinical trial designs must bridge this gap by moving beyond short-term surrogate biomarker endpoints to long-term outcome-driven randomized trials [1,29,30].

8. Conclusions

1. Intermittent fasting-most notably time-restricted eating-serves as an effective non-pharmacological strategy for improving cardiometabolic health, producing meaningful reductions in blood pressure, atherogenic lipid fractions, and insulin resistance.
2. The cardiovascular benefits of IF extend beyond simple weight loss; they are driven by metabolic switching, systemic autophagy upregulation, reduced vascular inflammation, and improved endothelial function.
3. Early time-restricted eating (eTRE) aligned with endogenous circadian clocks yields superior metabolic responses compared to late-evening eating windows.
4. Clinical implementation must be strictly individualized. Highly restrictive eating windows (<8 hours) should be avoided, and intensive medical supervision is mandatory for patients with diabetes or severe cardiovascular multi-morbidity to prevent hypoglycemia and hemodynamic instability.

5. High-quality, long-term randomized clinical trials assessing hard cardiovascular endpoints (MACE) and patient-reported outcomes are required before intermittent fasting can be formally incorporated into core cardiovascular preventive guidelines.

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Declaration of the use of generative AI and AI-assisted technologies in the writing process

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